

PULKIT GARG

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AREA OF EXPERTISE

Backend Development · Scalable Systems · REST APIs · Data Structures & Algorithms · System Design · Agentic AI

EDUCATION

Thapar Institute of Engineering and Technology, Patiala

Bachelor of Engineering in Computer Science and Engineering

Aug 2023 – May 2027

CGPA: 8.71

PROJECTS

SynthSales – Agentic AI Sales CRM

 Live | 

Tech: FastAPI, PostgreSQL, SQLAlchemy 2.0, Alembic, APScheduler, Redis, Docker, Google APIs, Gemini API

Agentic B2B outreach backend that turns uploaded company CSVs into researched, scored, contactable prospects with verified emails, AI-drafted outreach, reply tracking, and meeting scheduling.

- Built a scalable FastAPI backend with **78** REST endpoints and **14** PostgreSQL models, owning it end-to-end from design to JWT auth and admin approvals.
- Engineered an **8-agent pipeline** for enrichment, scoring, contact discovery, email verification, outreach, tracking, meetings, and reply classification.
- Engineered for reliability at scale: outbound kill-switches, do-not-contact suppression, **24h** undo snapshots, and Redis-backed rate limiting.

Real-Time Self-Healing Data Pipeline



Tech: Apache Airflow, Python, PostgreSQL, dbt, Docker, Streamlit, Ollama LLaMA 3.2, pytest

Fault-tolerant pipeline that detects schema drift, auto-repairs malformed CSVs, and monitors health end-to-end.

- Developed a 4-task Airflow pipeline for CSV validation, PostgreSQL ingestion, and dbt transformation.
- Implemented LLaMA 3.2 self-healing to repair schema drift and log every AI intervention in PostgreSQL.
- Wrote pytest suites and containerized the stack with Docker for reproducible, reliable deployments.

VOCA – Vocabulary Learning Platform

 Live | 

Tech: Next.js 14, MongoDB, React, LangChain, NextAuth.js, Tailwind CSS

Full-stack vocabulary learning platform for storing, organizing, and practicing English words with spaced repetition, progress tracking, and AI-assisted support.

- Constructed **14** authenticated API routes and **5** protected dashboard modules for word management, practice tracking, settings, and AI workflows.
- Designed a **6-level** spaced repetition system with **4** practice modes for reviews, streaks, and mastery tracking.
- Created a LangChain chatbot with persistent MongoDB memory, supporting **30** recent threads and replaying up to **50** messages per thread with one-click word saving.

TECHNICAL SKILLS

Languages & CS: C, C++, Python, JavaScript, Data Structures & Algorithms

Backend & APIs: FastAPI, REST APIs, Redis, NextAuth.js

Databases: SQL (MySQL, PostgreSQL), MongoDB, ChromaDB

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

AI & Automation: GenAI, LangChain, LangGraph, LLM Integration, Ollama, Streamlit

Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, Regression & Classification, Neural Networks, NLP

Cloud & DevOps: Git, GitHub, Docker, AWS (EC2, S3)

CERTIFICATIONS & ACHIEVEMENTS

Amazon ML Summer School

2026

Selected nationwide for Amazon's competitive ML training program

Advanced Learning Algorithms

Dec 2025

DeepLearning.AI & Stanford University (Coursera)

Supervised Machine Learning: Regression and Classification

Jan 2025

DeepLearning.AI & Stanford University (Coursera)